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ORIGINAL ARTICLE

Detection of Hypertrophic Cardiomyopathy on Electrocardiogram Using Artificial Intelligence

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BACKGROUND: Hypertrophic cardiomyopathy (HCM) is associated with significant morbidity and mortality, including sudden cardiac death in the young. Its prevalence is estimated to be 1 in 500, although many people are undiagnosed. The ability to screen electrocardiograms for its presence could improve detection and enable earlier diagnosis. This study evaluated the accuracy of an artificial intelligence device (viz HCM) in detecting HCM based on a 12-lead electrocardiogram.

METHODS: The device was previously trained using deep learning and provides a binary outcome (HCM suspected or not suspected). This study included 293 HCM-positive and 2912 HCM-negative cases, which were selected from 3 hospitals based on chart review incorporating billing diagnostic codes, cardiac imaging, and electrocardiogram features. The device produced an output for 291 (99.3%) HCM-positive and 2905 (99.8%) HCM-negative cases.

RESULTS: The device identified HCM with sensitivity of 68.4% (95% CI, 62.8–73.5%), specificity of 99.1% (95% CI, 98.7–99.4%), and area under the curve of 0.975 (95% CI, 0.965–0.982). With assumed population prevalence of 0.002 (1 in 500), the positive predictive value was 13.7% (95% CI, 10.1–19.9%) and the negative predictive value was 99.9% (95% CI, 99.9–99.9%). The device demonstrated broadly consistent performance across demographic and technical subgroups.

CONCLUSIONS: The device identified HCM based on a 12-lead electrocardiogram with good performance. Coupled with clinical expertise, it has the potential to augment HCM detection and diagnosis.

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Key Words: artificial intelligence ■ deep learning ■ electrocardiography ■ hospitals ■ prevalence

Hypertrophic cardiomyopathy (HCM) is a complex myocardial disorder that may result in substantial morbidity and mortality, including advanced heart failure and sudden cardiac death.[1] Its prevalence is estimated to be 1 in 500, although many people remain undiagnosed.[2–6] The ability to screen electrocardiograms (ECGs) for its presence could improve detection and enable earlier diagnosis. Given that HCM is often genetic, this approach has the potential to enhance the care of both individual patients and families.

A key challenge in diagnosing HCM is that it relies on a dichotomous measure of left ventricular wall thickness (threshold slightly modified based on family history), after excluding other potential factors that may increase wall thickness, particularly hypertension and storage or infiltrative disorders.[7] ECGs may show findings consistent with HCM, such as left ventricular hypertrophy, although such findings are not specific to HCM.[8] It has been proposed that an ECG artificial intelligence (AI) algorithm may be able to detect features that are more specific

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WHAT IS NEW?

- This retrospective study assessed the ability of an artificial intelligence device (viz hypertrophic cardiomyopathy) to detect hypertrophic cardiomyopathy based on a 12-lead ECG.
- The device achieved a sensitivity of 68.4% and specificity of 99.1%, which corresponded to a positive predictive value of 13.7% and a negative predictive value of 99.9%.
- The device demonstrated largely consistent performance across demographic and technical subgroups.

WHAT ARE THE CLINICAL IMPLICATIONS?

- Artificial intelligence will likely play a greater role in clinical care, including through enhancing the detection of conditions like hypertrophic cardiomyopathy.
- This device could be used for initial screening for hypertrophic cardiomyopathy, especially given its lower cost and complexity than echocardiography-based assessment.
- The performance across subgroups indicates the generalizability of the device, although its performance should continue to be evaluated as it is deployed clinically.
- Following this study, this device was granted a de novo request by the Food and Drug Administration as the first cardiovascular machine learning-based notification software device.

Nonstandard Abbreviations and Acronyms

AI	artificial intelligence
AUC	area under the curve
ECG	electrocardiogram
HCM	hypertrophic cardiomyopathy
ICD-9	International Classification of Diseases-Ninth Revision
ICD-10	International Classification of Diseases-Tenth Revision
MGB	Mass General Brigham
PPV	positive predictive value

to HCM.[9] Such an algorithm could facilitate screening patients for HCM, triggering further clinical evaluation when appropriate, to confirm the diagnosis.

This study investigated the viz HCM software-as-a-medical-device, which utilizes ECG as its input and has subsequently been granted de novo approval through the United States Food and Drug Administration.[10] It aimed to evaluate the performance of the device by comparing its outputs to ground truth interpretations as determined by expert clinical review. To assess robustness and generalizability, it also explored the performance in subgroups based on patient demographics, technical parameters, potentially confounding ECG findings, and coexisting diagnoses (including mimics of HCM).

METHODS

Study Design

This retrospective standalone model performance study was conducted using ECG cases from adult patients (aged ≥18 years) at 3 hospitals within the Mass General Brigham (MGB) network between July 1, 2017, and June 30, 2022. These 3 hospitals had not provided any training data for the model. The study was approved by the MGB institutional review board with a waiver of informed consent. It was conducted in accordance with relevant guidelines and regulations, including the Health Insurance Portability and Accountability Act. This report followed the Standards for Reporting Diagnostic Accuracy (STARD 2015) reporting guideline. The data that support the findings of this study are available from the corresponding author upon reasonable request.

Cohort Selection Process

This study included 300 HCM-positive and 3000 HCM-negative ECG cases from distinct patients. These cases were identified from a list of candidate cases in a consecutive manner until the required cohort size was achieved. The cohort selection occurred through a clinical record search based on *International Classification of Diseases-Ninth Revision* (ICD-9) and *International Classification of Diseases-Tenth Revision* (ICD-10) codes that was confirmed through a manual chart review. The cohort was essentially finalized after these processes, with a small number of cases being excluded during the subsequent ECG annotations and model inference (Figure 1).

The clinical record search for HCM-positive cases identified patients with the ICD-9 and ICD-10 codes for HCM (ICD-9: 425.1 Hypertrophic cardiomyopathy; ICD-10: I42.1 Obstructive hypertrophic cardiomyopathy or I42.2 Other hypertrophic cardiomyopathy) through the Research Patient Data Registry[11] at MGB. The clinical record search for HCM-negative cases identified patients without these ICD-9 and ICD-10 codes. Given the large number of available ECG cases to consider for the HCM-negative cases, ECGs were considered from 5 random months for each hospital (this strategy amounted to 200 cases for each hospital from each month). Both searches also excluded patients who had an ICD-9 or ICD-10 code for a pacemaker (ICD-9: V45.01; ICD-10: Z95.0), which was an exclusion criterion due to the alteration of ECGs in the presence of paced rhythms. There was no restriction on patient setting within the hospital (ie, ECGs were considered from emergency patients, inpatients, and outpatients).

The identified cases were then placed in a randomized order so that confirmatory chart reviews could follow the principles of consecutive cohort selection based on this order until sufficient cases were selected. These chart reviews were performed by board-certified cardiologists who were assisted by resident physicians and critical care nurses for components not requiring cardiology knowledge (eg, a patient not having had an imaging study). All HCM-positive cases and ≈55% of HCM-negative cases included in the final cohort were reviewed

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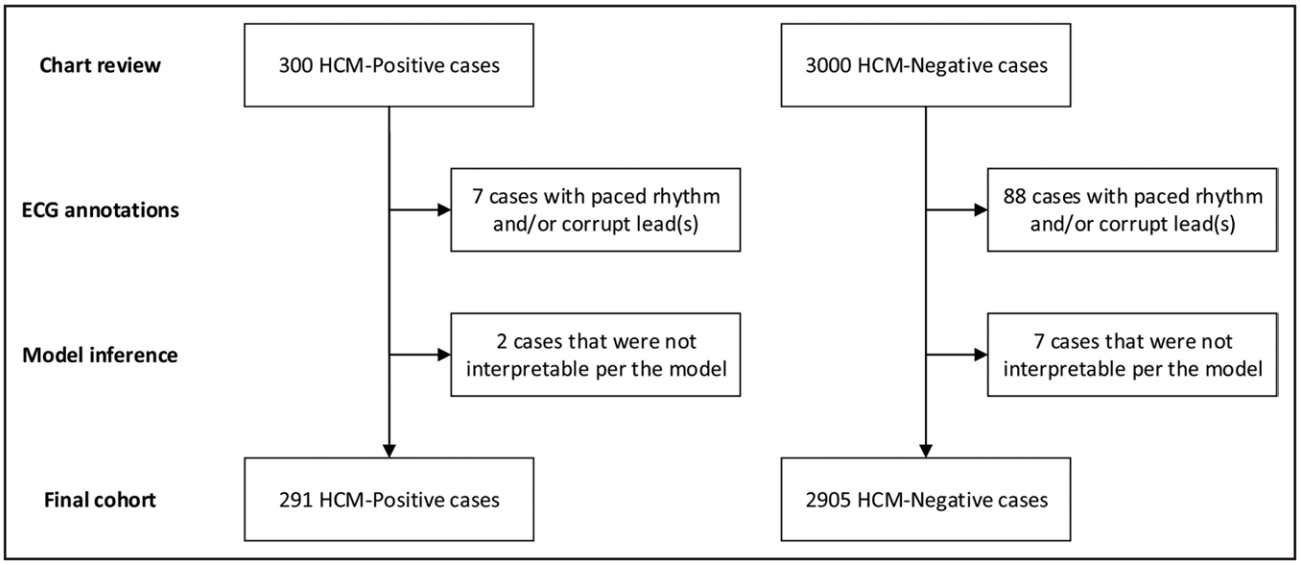

Figure 1. Cohort selection flowchart.
HCM indicates hypertrophic cardiomyopathy.

by a board-certified cardiologist. This information was entered directly into the REDCap electronic data capture tool hosted at MGB.[12,13] Once the HCM-positive and HCM-negative cohorts were finalized, the ECG cases underwent deidentification for use in subsequent steps.

HCM-Positive Case Chart Review

A manual chart review confirmed the patient's diagnosis of HCM, including whether it was obstructive or nonobstructive HCM. The ECG used for analysis was selected by taking the first ECG of the day on the date closest to the first echocardiography and cardiac magnetic resonance imaging (CMR) study that established the diagnosis of HCM. Cases were excluded if the patient was <18 years of age, had no confirmatory echocardiography and CMR, had an alternate diagnosis to HCM, had received septal myectomy or alcohol ablation before the ECG, had no evidence of left ventricular hypertrophy by both the Cornell and Sokolow-Lyon voltage criteria (ie, cases were included if either criterion was met), or had pacemaker presence and corrupt lead(s). The final cohort reflected a yield of ≈13% from the clinical record search using ICD-9/ICD-10 codes, with the most common reason for exclusion being that the patients had no confirmatory echocardiography and CMR for HCM (≈56% of cases from the clinical record search). It is noted that many of these cases did not have HCM (ie, as opposed to being excluded due to a lack of imaging). The results of genetic testing, left ventricular wall thickness, and left ventricular ejection fraction were also obtained from each patient's chart when available; the left ventricular wall thickness was based on the reported intraventricular septum thickness on echocardiography or the maximal wall thickness when an apical variant; the left ventricular wall thickness was based on the reported maximal wall thickness on CMR.

HCM-Negative Case Chart Review

A manual chart review confirmed the patient's absence of a diagnosis of HCM and identified the ECG used for analysis, which was the first one within the selected month at that hospital. The initial chart review involved searching for the terms HCM and hypertrophic cardiomyopathy, determining if the patient had had an echocardiography and CMR study within the study date range, and ensuring the selected ECG did not have pacemaker presence and corrupt lead(s). If any of these criteria were met then a cardiologist reviewed the case to ensure the patient did not have a diagnosis of HCM, the ECG did not have pacemaker presence, and the ECG did not have corrupt lead(s) before it was included in the final cohort. If none of the exclusion criteria were met, the case was included in the final cohort.

ECG Annotations

The ECG cases were reviewed by a cardiologist for the presence of 16 different findings or abnormalities: pacemaker presence, atrial fibrillation, premature atrial contraction(s)/premature ventricular contraction(s), left axis deviation, atrial enlargement, left ventricular hypertrophy, right ventricular hypertrophy, increased precordial voltages, acute ischemic change(s), myocardial infarct of indeterminate age, nonspecific ST segment change(s), nonspecific T wave abnormality, ECG containing a corrupt lead(s), other ECG artifact or noise, other pathological finding (with free text), and no pathological finding. This step incorporated mixed HCM-positive and HCM-negative cases such that the ECG annotations would not be influenced by the knowledge of a case being positive or negative. If an ECG was annotated as having pacemaker presence or containing a corrupt lead(s), it was excluded. These annotations were performed using the Food and Drug Administration-cleared eUnity image visualization software (Version 6 or higher) and a previously described internal web-based annotation system that utilized REDCap.[12–14]

Comorbid Conditions

The clinical record search also returned information about the presence of comorbid conditions based on ICD-9 and ICD-10 codes. The presence or absence of the following

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cardiopulmonary conditions was recorded: ischemic heart disease (ICD-9: 410-414; ICD-10: I20-I25), pulmonary hypertension (ICD-9: 416.0; ICD-10: I27.0, I27.2), pulmonary heart disease (ICD-9: 416.8, 416.9; ICD-10: I27.8, I27.9), nonrheumatic aortic stenosis (ICD-9: 424.1; ICD-10: I35.0, I35.2), left bundle branch block, unspecified (ICD-9: 426.2, 426.3; ICD-10: I44.7), right bundle branch block (ICD-9: 426.4, 426.5; ICD-10: I45.0, I45.1, I45.2, I45.3), preexcitation syndrome (which incorporates Wolf-Parkinson-White syndrome; ICD-9: 426.7; ICD-10: I45.6), cardiomegaly (which incorporates Athlete's heart; ICD-9: 429.3; ICD-10: I51.7), congenital heart disease (ICD-9: 745-746; ICD-10: Q20-Q24), and coarctation of the aorta (ICD-9: 747.1; ICD-10: Q25.1); hypertension (ICD-10: I10-I16 and I1A) was added as a post hoc exploratory analysis. The presence or absence of the following systemic conditions was recorded: sarcoidosis (ICD-9: 135; ICD-10: D86), glycogen storage disease (ICD-9: 271.0; ICD-10: E74.0), Fabry (-Anderson) disease (ICD-9: 272.7; ICD-10: E75.21), hemochromatosis (ICD-9: 275.01, 275.02, 275.03; ICD-10: E83.11), amyloidosis (ICD-9: 277.3; ICD-10: E85), and Friedreich ataxia (ICD-9: 334.0; ICD-10: G11.11). However, no patients with Friedreich ataxia were identified within the cohorts based on the ICD-9/ICD-10 codes.

Model Inference

The evaluated AI model was version 1.0 of the viz HCM device, which was developed separately for this article using proprietary methods. This device initially assessed the ECG data for the appropriate format to determine if the source file was supported and contained the essential components that are required to inspect the ECG metadata and signal quality. It then inspected the metadata of the ECG source file to assess eligibility of the patient ECG data for analysis. It also inspected the ECG signal for the presence of a pacemaker, in which case it would not provide an interpretation. It then performed a quality check of the ECG signal to identify ECGs with a suboptimal signal (ie, corrupt or missing leads). It then preprocessed and converted the ECG signal into different tensors, which were fed into an ensemble of neural network models for inference.

The device output consisted of an HCM classification score between 0 and 1 (with scores closer to 1 indicating a greater likelihood of HCM) and a binary output (HCM suspected or not suspected) above a sponsor-defined threshold. It also output the inference time. The model was installed at MGB for use in this study and received only the deidentified ECG waveform file.

Statistical Analysis

The statistical analysis was performed in R (version 4.0.2). The predefined primary end points were the sensitivity and specificity for suspected HCM detection. The predefined secondary end point was the positive predictive value (PPV). The predefined secondary assessments included the area under the receiver operating characteristic curve (AUC), negative predictive value, and descriptive summaries of the model inference time. They also included subgroup analyses of sensitivity for HCM characterization (ie, nonobstructive versus obstructive), and sensitivity and specificity for patient demographics (age range, sex, race, ethnicity), technical parameters (ECG machine model), hospital site, presence of coexisting ECG findings or abnormalities, and the presence of chart-based coexisting conditions. Post hoc exploratory analyses included subgroup analyses of sensitivity based on patient sarcomere variant status, left ventricular wall thickness, and left ventricular ejection fraction. The patient demographic and technical parameter data were derived from clinical databases and ECG file metadata, respectively; any missing data were treated as unknown and no data were imputed.

The CIs for sensitivity and specificity were calculated using the Agresti-Coull method, which allows sensitivities and specificities to be <0% or >100%. The CIs for PPV, negative predictive value, and AUC were calculated using bootstrapping with 2000 intervals. Formal comparison testing was not performed between subgroups, with the 95% CIs being the focus for evaluating any differences. The sample sizes were powered based on preliminary model results to achieve a sensitivity and specificity that would enable a PPV ≥5.0% using a prevalence of 0.002 (1 in 500).

RESULTS

Cohort Selection

The cohort selection process yielded 300 HCM-positive and 3000 HCM-negative cases (Figure 1). The subsequent ECG annotations showed that these cases inadvertently included 7 (2.3%) HCM-positive and 88 (2.9%) HCM-negative cases with paced rhythm and corrupt lead(s), and they were removed from the cohort. The model also identified 2 (0.7%) HCM-positive and 7 (0.2%) HCM-negative cases that were noncompliant based on its quality control checks. Model inference was performed on the remaining 291 HCM-positive and 2905 HCM-negative cases.

The HCM-positive cases included 149 (51.2%) nonobstructive and 142 (48.8%) obstructive HCM cases (Table 1). These cases came from patients with a mean age (±SD) of 61.3±16.8 years and 139 (47.8%) female patients; 15 (5.2%) patients had a sarcomere variant, 52 (17.9%) had no sarcomere variant, and 224 (77.0%) had an unknown sarcomere variant status. The HCM-negative cases came from patients with a mean age (±SD) of 59.8±18.0 years and 1501 (51.7%) female patients. The race, ethnicity, and ECG machine model baseline information for HCM-positive and HCM-negative cases are provided in Table 1.

Model Performance

The model identified HCM with a sensitivity of 68.4% (95% CI, 62.8–73.5%; 199 true positive cases and 92 false negative cases) and specificity of 99.1% (95% CI, 98.7–99.4%; 2880 true negative cases and 25 false positive cases; Table 2). Assuming a prevalence of 1:500, these results corresponded to a PPV of 13.7% (95% CI, 10.1–19.9%) and a negative predictive value of 99.9% (95% CI, 99.9–99.9%). The model had an AUC of 0.975 (95% CI, 0.965–0.982; Figure 2). There

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Table 1. Baseline Demographic and Technical Information for ECG Cases

	HCM-positive			
	Nonobstructive	Obstructive	Total	HCM-negative
No. of cases	149	142	291	2905
Sarcomere variant status				
Present	10 (6.7%)	5 (3.5%)	15 (5.2%)	NA
Absent	24 (16.1%)	28 (19.7%)	52 (17.9%)	NA
Unknown	115 (77.2%)	109 (76.8%)	224 (77.0%)	NA
Gender				
Female	61 (40.9%)	78 (54.9%)	139 (47.8%)	1501 (51.7%)
Male	88 (59.1%)	64 (45.1%)	152 (52.2%)	1404 (48.3%)
Age				
Mean±SD (y)	58.4±18.0	64.2±15.1	61.3±16.8	59.8±18.0
≥18 to <40 y	27 (18.1%)	11 (7.7%)	38 (13.1%)	486 (16.7%)
≥40 to ≤65 y	68 (45.6%)	63 (44.4%)	131 (45.0%)	1178 (40.6%)
>65 y	54 (36.2%)	68 (47.9%)	122 (41.9%)	1241 (42.7%)
Race				
American Indian or Alaska Native	0 (0.0%)	0 (0.0%)	0 (0.0%)	2 (0.1%)
Asian	12 (8.1%)	3 (2.1%)	15 (5.2%)	94 (3.2%)
Black	22 (14.8%)	9 (6.3%)	31 (10.7%)	264 (9.1%)
Native Hawaiian or Other Pacific Islander	0 (0.0%)	0 (0.0%)	0 (0.0%)	1 (0.0%)
White	109 (73.2%)	114 (80.3%)	223 (76.6%)	2243 (77.2%)
Other	3 (2.0%)	6 (4.2%)	9 (3.1%)	193 (6.6%)
Two or more	2 (1.3%)	2 (1.4%)	4 (1.4%)	21 (0.7%)
Declined	1 (0.7%)	3 (2.1%)	4 (1.4%)	34 (1.2%)
Unknown	0 (0.0%)	5 (3.5%)	5 (1.7%)	53 (1.8%)
Ethnicity				
Hispanic	3 (2.0%)	4 (2.8%)	7 (2.4%)	170 (5.9%)
Non-Hispanic	126 (84.6%)	120 (84.5%)	246 (84.5%)	2506 (86.3%)
Declined	6 (4.0%)	11 (7.7%)	17 (5.8%)	120 (4.1%)
Unknown	14 (9.4%)	7 (4.9%)	21 (7.2%)	109 (3.8%)
ECG machine model				
CASE	0 (0.0%)	0 (0.0%)	0 (0.0%)	2 (0.1%)
D3K	17 (11.4%)	12 (8.5%)	29 (10.0%)	542 (18.7%)
MAC16	0 (0.0%)	0 (0.0%)	0 (0.0%)	3 (0.1%)
MAC2K	10 (6.7%)	17 (12.0%)	27 (9.3%)	165 (5.7%)
MAC55	116 (77.9%)	108 (76.1%)	224 (77.0%)	2028 (69.8%)
MAC5K	0 (0.0%)	0 (0.0%)	0 (0.0%)	1 (0.0%)
MV360	1 (0.7%)	0 (0.0%)	1 (0.3%)	37 (1.3%)
S8500	5 (3.4%)	5 (3.5%)	10 (3.4%)	127 (4.4%)

HCM indicates hypertrophic cardiomyopathy.

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were no notable differences in sensitivity (a difference was considered notable when the 95% CIs did not overlap) for obstructive versus nonobstructive HCM, sarcomere variant status, left ventricular wall thickness, and left ventricular ejection fraction (Table 2; Figure 3). The model had a mean processing time (±SD) of 1.8±0.3 seconds; the range was 0.8 to 7.8 seconds. For example, ECGs for different model accuracies are provided in Figure 4.

Demographic and Technical Subgroup Performance

The model identified HCM with an AUC >0.970 for all gender, age, race, ethnicity, ECG machine model, and hospital site subgroups (Table S1) except for the unknown ethnicity subgroup (AUC, 0.964) and hospital site 3 (AUC, 0.948). There were no demographic or technical subgroups where there was a notable

Table 2. Model Performance Summary

Overall	
AUC	0.975 (0.965–0.982)
Sensitivity	68.4% (62.8–73.5%)
Specificity	99.1% (98.7–99.4%)
PPV	13.7% (10.1–19.9%)
NPV	99.9% (99.9–99.9%)
Sensitivity for subgroups	
HCM characterization	
Nonobstructive cases (n=149)	68.5% (60.6–75.4%)
Obstructive cases (n=142)	68.3% (60.2–75.4%)
Sarcomere variant status	
Present (n=15)	73.3% (47.6–89.5%)
Absent (n=52)	75.0% (61.7–84.9%)
Unknown (n=224)	66.5% (60.1–72.4%)
Left ventricular wall thickness	
<13 mm (n=47)	74.5% (60.4–84.9%)
≥13 mm (n=233)	67.4% (61.1–73.1%)
Left ventricular ejection fraction	
<50% (n=10)	60.0% (31.2–83.3%)
≥50% (n=278)	68.7% (63.0–73.9%)

This summary includes the 95% CIs in parentheses. There were 11 cases without a reported left ventricular wall thickness and 3 cases without a reported left ventricular ejection fraction. AUC indicates area under the curve; HCM, hypertrophic cardiomyopathy; n, number of cases; NPV, negative predictive value; and PPV, positive predictive value.

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difference in sensitivity or specificity (when the 95% CIs from subgroups did not overlap). However, the model did show a trend for lower sensitivity in cases from female patients (61.2% [95% CI, 52.8–68.9%]) compared with male patients (75.0% [95% CI, 67.5–81.2%]), yet higher specificity (99.5% [95% CI, 99.0–99.8%] in cases from female patients versus 98.7% [95% CI, 98.0–99.2%] from male patients). The model also showed a trend for higher sensitivity for cases from patients aged ≥18 to <40 years (84.2% [95% CI, 69.2–92.9%]) compared with patients aged ≥40 to ≤65 years (64.1% [95% CI, 55.6–71.8%]) or >65 years (68.0% [95% CI, 59.3–75.7%]). Among the 3 most common racial groups in the cohort, the model had a trend for higher sensitivity for cases from Asian patients (73.3% [95% CI, 47.6–89.5%]) compared with Black patients (64.5% [95% CI, 46.9–79.0%]) and White patients (66.8% [95% CI, 60.4–72.7%]); the specificities were similar across these 3 racial groups (98.9% [95% CI, 93.6–100.4%] for cases from Asian patients, 99.2% [95% CI, 97.1–100.0%] from Black patients, and 99.1% [95% CI, 98.6–99.4%] from White patients).

Other ECG Findings Subgroup Performance

The model was also assessed for its ability to identify HCM in the presence or absence of other ECG findings (Table S2). For many ECG findings, the performance was similar in the presence or absence of each finding. The findings where there was a notable difference in sensitivity (when the 95% CIs did not overlap) included atrial fibrillation (46.2% [95% CI, 28.7–64.5%] when present versus 70.6% [95% CI, 64.8–75.7%] when absent), left ventricular hypertrophy (74.7% [95% CI, 68.7–79.9%] when present versus 45.2% [95% CI, 33.4–57.5%] when absent), nonspecific ST segment changes (76.0% [95% CI, 68.5–82.2%] when present versus 60.3% [95% CI, 52.0–68.0%] when absent), and no pathological finding (25.0% [95% CI, 6.3–59.9%] when present versus 69.6% [95% CI, 64.0–74.7%] when absent). The findings where there was a notable difference in specificity included left ventricular hypertrophy (95.8% [95% CI, 92.6–97.7%] when present versus 99.5% [95% CI, 99.1–99.7%] when absent) and nonspecific ST segment changes (97.7% [95% CI, 95.5–98.8%] when present versus 99.4% [95% CI, 99.0–99.6%] when absent).

Presence of Comorbid Conditions Subgroup Performance

The model was assessed for its ability to identify HCM in the presence or absence of comorbid conditions, including cardiopulmonary and systemic conditions (Table S3). For many comorbid conditions, the performance was similar in the presence or absence of each finding. There were no comorbid conditions with a notable difference in sensitivity (when the 95% CIs did not overlap). The comorbid conditions where there was a notable difference in specificity included nonrheumatic aortic stenosis (96.2% [95% CI, 93.1–98.0%] when present versus 99.4% [95% CI, 99.0–99.7%] when absent) and congenital heart disease (97.1% [95% CI, 93.1–98.9%] when present versus 99.3% [95% CI, 98.9–99.5%] when absent).

DISCUSSION

This study assessed the ability of an AI model to identify people with suspected HCM based on their ECG. This model overall performed well, achieving a sensitivity of 68.4% and specificity of 99.1%. These metrics corresponded to a PPV of 13.7%, a negative predictive value of 99.9%, and an AUC of 0.975. Based on these results, this model was granted a de novo request by the Food and Drug Administration as the first device under regulation number 21 CFR 870.2380 for cardiovascular machine learning-based notification software.[10]

A key consideration for the current algorithm was ensuring an adequate balance between identifying true positive and false positive cases, as reflected by the PPV. The operating point for this device was therefore chosen to have a higher specificity compared with sensitivity. Two

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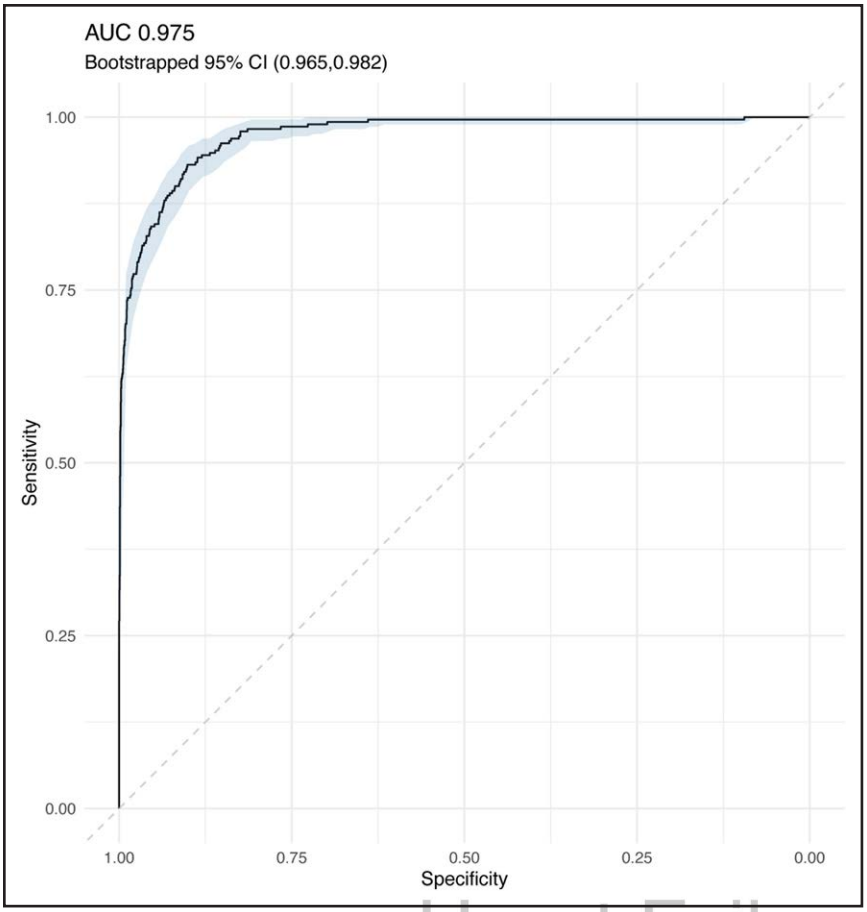

Figure 2. Model performance. This receiver operating characteristic curve includes the 95% CI in shading. AUC indicates area under the curve.

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algorithms in the research setting have demonstrated a similar AUC; they have utilized operating points that balance sensitivity and specificity differently, which we note would correspond to a lower PPV. An algorithm developed at the Mayo Clinic performed with AUC 0.96, sensitivity 87%, and specificity 90%.[15] A separate algorithm developed at Yale New Haven Hospital had AUC 0.96, sensitivity 91%, and specificity 91%.[16]

The clinical utility of an ECG-based diagnostic device in initial screening for HCM is that it can be implemented

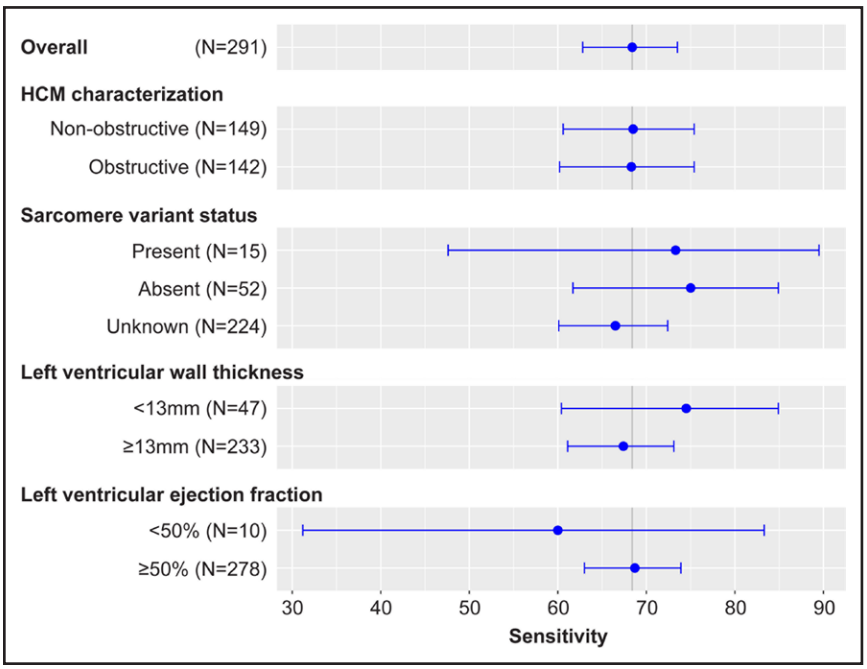

Figure 3. Sensitivity for subgroups. This forest plot demonstrates the sensitivity for different subgroups of HCM-positive cases. The blue dots reflect the point estimates, the blue brackets reflect the 95% CIs and the gray vertical line reflects the overall sensitivity point estimate. HCM indicates hypertrophic cardiomyopathy; and N, number of cases.

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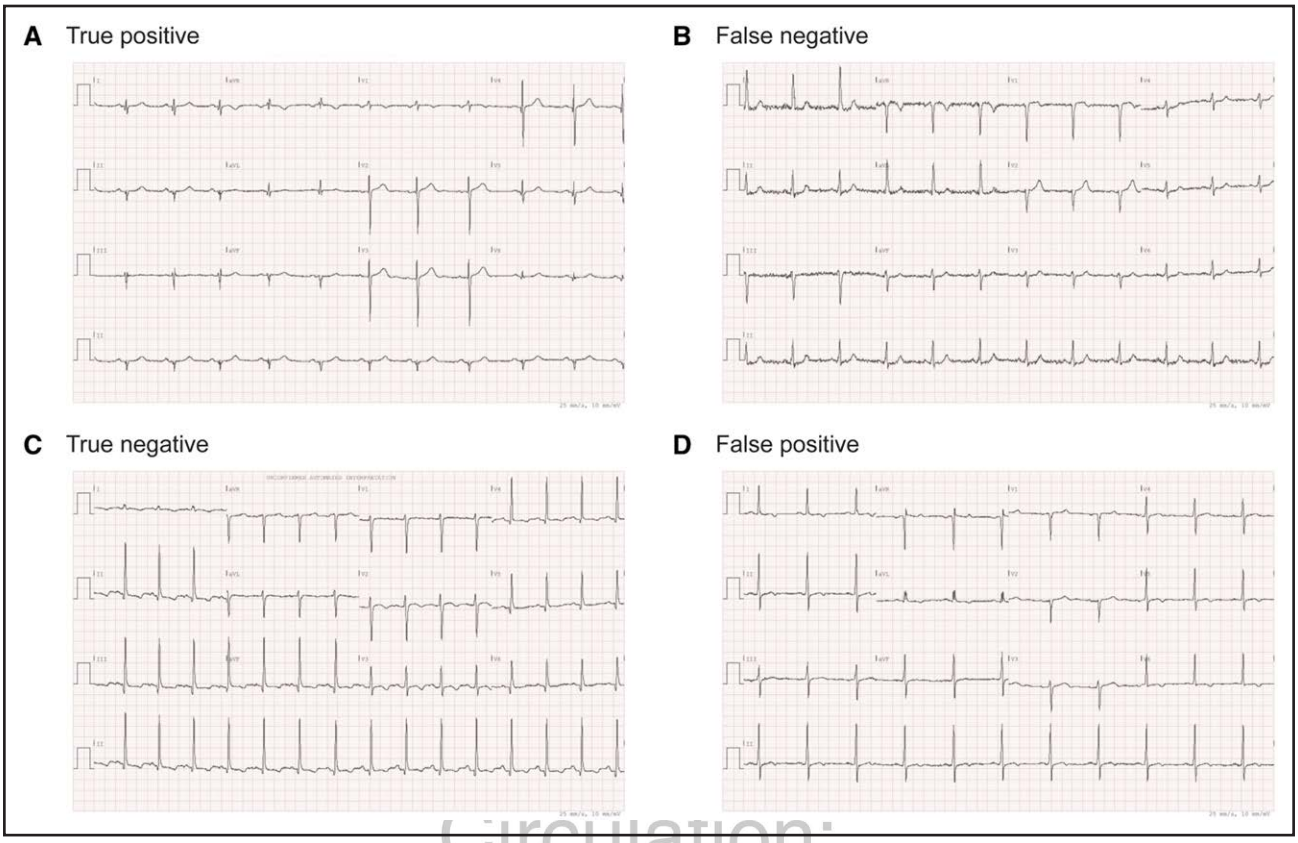

Figure 4. Example ECG cases.
These cases demonstrate HCM-positive cases that the model correctly identified as suspected HCM (**A**) or did not correctly identify (**B**), and HCM-negative cases that the model correctly identified as not suspected (**C**) or did not correctly identify (**D**).

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at lower cost and complexity than echocardiography-based assessment. The specific application may vary from one setting to another. An example is the large-scale screening of young, mostly healthy individuals before participation in physical activities such as athletes and military recruits; the sensitivity of 84.2% among the ≥18 to <40 years subgroup is particularly appealing in this setting. Given the ability to function automatically, the device could separately analyze all ECGs obtained in defined clinical settings to identify patients who could benefit from cardiology evaluation.

The device demonstrated broadly consistent performance across different demographic groups. A notable difference was the trend for higher sensitivity among ECGs obtained from male compared with female patients, which may relate to the larger voltages typically seen on ECGs from male patients. This study tested the current Food and Drug Administration-approved version of this device, which utilizes an operating point that is akin to a normal threshold value for a laboratory test. Future device iterations may consider a lower operating point for women.

The race of patients is known to impact their ECG.[17] This study had included over 100 ECG cases (≈10% of which were HCM-positive) from patients of each of Asian, Black, and White racial backgrounds. The sensitivity was similar between patients from the Black and White subgroups, while patients from the Asian subgroup had a trend for improved sensitivity. The cause for this increased sensitivity is not apparent, although echocardiographic differences, including more frequent apical hypertrophy, have been described in Asian compared with European health care centers.[18] The consistency across these racial groups provides reassurance about device generalizability. Its ongoing ability to perform for people from different racial subgroups, especially those with smaller representation in this cohort, should continue to be assessed.

A limitation of this study included that it was performed in a retrospective manner on cases where a diagnosis of HCM had already been made, while its potential use is in a prospective manner on cases where a diagnosis of HCM has not been made. This restriction was in large part because this assessment of the AI model aimed to show that it has the accuracy and therefore potential to be beneficial. The key clinical effectiveness question moving forward will be how many cases of undiagnosed HCM it helps identify; a recent study about the same model showed that 5% of the patients flagged for suspected HCM had a new

diagnosis of HCM.[19] Similarly, the HCM-negative cases may reflect cases where there was an indication for an ECG and do not reflect the full spectrum of negative cases among the general population. Further assessments of its clinical effectiveness, as well as its performance in settings beyond those tested as part of this assessment, will be required.

The study design required HCM-positive cases to demonstrate ECG criteria for left ventricular hypertrophy during initial selection. ECGs were assessed again for left ventricular hypertrophy at the time of ECG annotations, which was one of 16 different ECG findings. During these ECG annotations, a proportion of HCM-positive cases were annotated as not containing left ventricular hypertrophy (ie, were HCM-positive left ventricular hypertrophy-absent). In a review of such cases, many appeared to have borderline left ventricular hypertrophy. We were reassured that the model still performed with sensitivity of 45.2% in cases with this discrepancy. Similarly, we were reassured that the model maintained specificity of 95.8% on the HCM-negative left ventricular hypertrophy-present cases, showing that it has the ability to discriminate between the presence or absence of HCM in spite of the presence of left ventricular hypertrophy.

The need for HCM-positive studies to demonstrate left ventricular hypertrophy may have also resulted in cases that had stronger evidence for HCM on ECG. This criterion reflected an initial regulatory body request to assess the model performance on cases that had a corresponding abnormality on the ECG (ie, through having left ventricular hypertrophy). It is recognized that many people with HCM do not have left ventricular hypertrophy on ECG. As explained in the prior paragraph, it appears that the model successfully identifies HCM in many cases with borderline left ventricular hypertrophy. Future research should evaluate the extent to which HCM-positive cases without left ventricular hypertrophy on ECG can be identified. Other future research questions include whether the model can distinguish between genotype-positive and genotype-negative patients and whether it can differentiate between phenotype-positive and phenotype-negative patients among genotype-positive patients, especially noting the low rate of genetic testing among the current HCM-positive cases.

CONCLUSIONS

The ability to detect HCM on 12-lead ECG has the potential to assist with screening for HCM. We demonstrated the accuracy of an AI model in identifying suspected HCM on ECG, including its performance across a range of demographic and technical subgroups, and in the presence of other ECG findings or comorbid conditions.

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Supplemental Material

Tables S1–S3

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